

A Living Meta-Analysis Architecture for Active Inference

Assertion Extraction, Nanopublications, and Hypothesis Scoring

Daniel Friedman
Active Inference Institute
daniel@activeinference.institute
ORCID: 0000-0001-6232-9096
and Joel Dietz
Massachusetts Institute of Technology (MIT)
California Institute for Machine Consciousness (CIMC)
jdietz@mit.edu
ORCID: 0000-0002-9456-2691
DOI: 10.5281/zenodo.19461934

July 24, 2026

Contents

1 Abstract

No prior automated system tracks hypothesis-level evidence mapping across the full Active Inference and Free Energy Principle (FEP) literature. Manual synthesis cannot keep pace with a field that has grown at a compound annual rate of 20.36% across 2000–2026 (empirical publication span 2005–2026), and the FEP’s theoretical generality has invited falsifiability critiques that only hypothesis-specific triage can address. Building on pioneering systematic manual annotation paired with ontology-based analysis at the scale of hundreds of papers, we present a computational meta-analysis framework that automates and scales this approach. The pipeline retrieves literature from arXiv, Semantic Scholar, and OpenAlex, deduplicating $N = 817$ papers (inclusion from 2000 onward) via a canonical identifier hierarchy (DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID). It classifies papers into a three-tier taxonomy spanning eight categories: A (Core Theory), B (Tools & Translation), and C (Application Domains). An LLM-powered extraction system evaluates each abstract against eight core hypotheses, separating (1) explicit source claims, (2) whether the abstract supplies evidence, and (3) pipeline hypothesis triage—each encoded in structured nanopublications with full extraction provenance.

All extracted assertions are machine-generated; hypothesis scores report citation-weighted evidence mapping and triage, not scientific confirmation. In the absence of a human-annotated gold set, a stratified agreement study ($n = 200$) compares pipeline triage against an independent, deterministic keyword-rule reference: pipeline precision 0.000 and recall 0.000 against the rule reference, with reference–pipeline direction agreement of only $\kappa = -0.055$. This low agreement is the point—absolute triage labels are unreliable, so we report relative rankings and trajectories rather than point estimates; human gold-standard validation remains future work.

The resulting evidence landscape reveals a field where application domains (Domain C, 64.0%) collectively dominate the corpus, with tools development (Domain B, 20.8%) and core theory (Domain A, 15.2%) rounding out the taxonomy. Non-negative matrix factorization identifies 8 latent topics that cross-cut the keyword taxonomy, and citation network analysis exposes a sparse yet structured graph (2,176 intra-corpus edges out of 29,323 total outgoing references—**only 7.4% reference resolution**) anchored by pronounced hub papers. Hypothesis triage scores cluster into tiers for relative comparison; absolute magnitudes are inflated by publication bias and linguistic asymmetry, so rankings and temporal trajectories are more reliable than point estimates. This work provides a reusable architecture for *living literature reviews*—continuously updated knowledge graphs that map citation-weighted evidence and support hypothesis triage across rapidly evolving fields.

Keywords: Active Inference, Free Energy Principle, meta-analysis, knowledge graph, nanopublications, bibliometrics, hypothesis scoring, LLM extraction, computational neuroscience

2 Introduction: Evidence Gaps in a Rapidly Expanding Field

2.1 The Free Energy Principle and Active Inference Framework

The Free Energy Principle (FEP), introduced by Karl Friston, proposes that self-organizing systems maintain their structural and functional integrity by minimizing variational free energy—an upper bound on sensory surprise (??). Under this principle, living systems are cast as approximate Bayesian inference engines that build generative models of their environment and act to reduce the discrepancy between predicted and observed states. Active Inference (AIF) extends this picture from passive perception to goal-directed behavior: agents select actions that bring about observations consistent with their preferred states, unifying perception, learning, and decision-making within a single variational framework (??). Since its initial formulation for sensorimotor control, AIF has been applied to navigation, visual foraging, language comprehension, social cognition, and multi-agent coordination. Bayesian mechanics (?) has further strengthened the mathematical foundations of the FEP by grounding Markov blanket dynamics in the physics of belief-based systems, placing the principle on a footing commensurate with established physical theories. Importantly, the variational free energy minimization at the core of the FEP shares deep mathematical connections with the broader family of Energy-Based Models (EBMs) (?)—including Helmholtz machines (?), Boltzmann machines (?), and variational autoencoders (?)—all of which parameterize learning and inference through scalar energy functions and variational bounds. This convergence motivates the inclusion of EBM-adjacent literature in our search scope.

2.2 Challenges Posed by Rapid Literature Growth

The active inference literature has grown at a compound annual rate of 20.36% across 2005–2026, with annual output accelerating sharply after 2013. While early research concentrated on theoretical neuroscience, the field has since diversified across biology (C5), robotics (C2), computational psychiatry (C4), algorithm scaling (B), and formal mathematics (A1). With $N = 817$ papers spanning 8 categories across 3 domains, no prior automated system tracks hypothesis-level evidence across the full corpus. This creates three interrelated challenges. First, the balance of evidence for core claims—such as FEP universality or the physical realism of Markov blankets—cannot be assessed without structured, hypothesis-specific extraction at corpus scale. Second, because the relationship between mathematical formalisms and empirical evidence is frequently implicit, systematic evidence synthesis demands substantial manual effort: Knight et al. (?) required human annotators to manually code hundreds of papers. Third, new entrants must navigate a literature weighted toward broad qualitative philosophy (A2), interspersed with specialized applied subfields whose findings are difficult to locate without domain-specific search strategies.

Traditional narrative reviews attempt to address these challenges but are static, subjective, and quickly outdated. Systematic reviews from evidence-based medicine offer rigorous aggregation but are structured for clinical trial data with homogeneous outcome measures, making them poorly suited for the heterogeneous ontological and computational claims in this literature. The expansion of predictive processing (??) and the emergence of Bayesian mechanics (?) further broaden the scope of assertions that a comprehensive meta-analysis must reconcile. Critically, the falsifiability of the FEP itself remains contested (?): because free energy minimization can be reframed to accommodate any behavior post hoc, distinguishing genuine predictive commitment from tautological redescription requires exactly the hypothesis-specific, evidence-quantified framework we propose here.

2.3 Related Work and Prior Meta-Analyses

Several prior efforts have surveyed aspects of the Active Inference landscape. Sajid et al. (?) compare active inference with alternative decision-making frameworks; Da Costa et al. (?) synthesize the discrete-state-space formulation; Lanillos et al. (?) survey robotics applications; Smith et al. (?) provide a tutorial bridging theory and empirical data; and Millidge et al. (?) examine information-theoretic foundations of exploration behavior. Ramstead et al. (?) extend the FEP to questions of biological self-organization, while Pezzulo et al. (?) connect active inference to homeostatic regulation. Millidge (?) provides a practitioner’s retrospective confirming that AIF’s strongest demonstrated results arise from novel discrete generative models, while scalability relative to deep reinforcement learning remains the field’s central open challenge.

Parallel to these synthesis efforts, Sanjeev V. Namjoshi’s 2026 textbook, *Fundamentals of Active Inference* (?), provides a comprehensive, computationally explicit foundation for the field designed for engineers. In conjunction with this text, Namjoshi developed the **aif-fep-db** repository (?)—an open-source, dynamically updated database of scraped and tagged publications covering active inference, the free energy principle, and predictive processing. While **aif-fep-db** curates and categorizes the literature to facilitate reproducible systematic reviews and interactive Dash-based exploration, it functions primarily as a modular bibliographic foundation rather than an automated hypothesis

evaluation engine.

Closest to our work, Knight, Cordes, and Friedman (?) conducted a systematic literature analysis of publications using the terms “Free Energy Principle” or “Active Inference,” with an emphasis on works by Karl J. Friston. Their analysis—maintained by the Active Inference Institute—combined manual annotation of structural, visual, and mathematical features with automated analyses using the Active Inference Ontology at the scale of thousands of citations and hundreds of annotated papers. That study identified six development directions—including broader scope, richer annotation, and transferable approaches—and represents an important precursor to automated meta-analysis of this field.

These prior works differ from the present study along four dimensions. First, **scale**: narrative reviews cover tens to low hundreds of papers; our pipeline processes $N = 817$. Second, **structure**: prior reviews produce prose summaries rather than machine-queryable knowledge graphs with typed relationships. Third, **temporal tracking**: no prior system computes how evidence for specific hypotheses evolves year over year. Fourth, **automation**: the systematic analysis of Knight et al. (?) pioneered quantitative literature analysis but relied on manual annotation, limiting update frequency. Our framework advances this line of work by (1) fully automating assertion extraction via LLM-based hypothesis scoring, (2) constructing a structured, RDF-compatible knowledge graph scored by citation-weighted evidence, and (3) tracking how evidence for core claims evolves over time through temporal trend analysis.

2.4 Synergizing Knowledge Graphs and LLMs

Broadening this synthesis, recent systematic literature initiatives underscore a powerful reciprocal synergy between Large Language Models (LLMs) and Knowledge Graphs: LLMs parse unstructured text to rapidly extract semantic claims, efficiently populating the structured, queryable architecture of the graph (??). We adopt the *nanopublication* (?)—a minimal, machine-readable unit of scientific evidence comprising a core assertion bound to explicit provenance metadata—as the fundamental serialization format for this extracted knowledge.

2.5 This Study: Approach and Overview

This paper presents a computational meta-analysis of the Active Inference literature ($N = 817$). Rather than relying exclusively on bibliometric metadata or slow manual coding, we deploy a Large Language Model (LLM) to “read” each paper’s abstract and assess its relationship to eight core hypotheses within the FEP paradigm. We serialize these assessments as nanopublications—each encoding an assertion (“Paper X supports Hypothesis Y”) coupled with the LLM’s natural-language reasoning and confidence score. The resulting knowledge graph aggregates these nanopublications and links them to paper metadata, citation networks, subfield classifications, and hypothesis definitions. A citation-weighted scoring formula quantifies the net evidence for or against each hypothesis, producing scores in $[-1, 1]$ that reflect both the direction and strength of published evidence. Importantly, this represents an open-source introductory analysis which will be augmented and extended, and stewarded in collaborative development by the Active Inference Institute (activeinference.org).

2.6 Research Questions

This meta-analysis addresses four primary research questions:

1. **RQ1 (Field Structure)**: What is the disciplinary structure and growth trajectory of the Active Inference literature, and how are papers distributed across the three domains—Core Theory (A), Tools & Translation (B), and Application Domains (C)? We expect Domain A to dominate but anticipate measurable diversification into applied domains.
2. **RQ2 (Growth Dynamics)**: What are the temporal growth dynamics of the field, and which subfields are experiencing the most rapid expansion? Prior reviews suggest accelerating growth post-2013; we quantify this trajectory and identify which application domains drive it.
3. **RQ3 (Hypothesis Evidence)**: What is the current balance of evidence for and against the eight standard hypotheses, and how has this balance evolved over time? We expect well-established hypotheses (H4 Predictive Coding) to show consensus while philosophically contested claims (H3 Markov Blanket Realism) show mixed evidence. (See hypothesis dashboard and assertion figures in the hypothesis results.)
4. **RQ4 (Tooling Readiness)**: What is the state of software tooling and infrastructure for Active Inference research, and what gaps remain? We survey available implementations to identify whether the ecosystem is fragmented or converging.

2.7 Scope and Delimitations

This study focuses on the English-language peer-reviewed and preprint literature retrievable from arXiv, Semantic Scholar, and OpenAlex. Our search scope begins at 2005—chosen to capture Energy-Based Model and variational Bayesian antecedents (Helmholtz machines, VAEs, early Bayesian brain formulations (??)) that share deep mathematical foundations with variational free energy minimization and predated the Free Energy Principle label introduced in 2006 (?). The scope includes both the core Active Inference and Free Energy Principle literature and adjacent Energy-Based Model research where it intersects with variational inference or generative modeling—capturing the growing convergence between these traditions. We do not include book chapters or monographs not indexed by these sources, software documentation, or non-English publications. Domain classification uses keyword matching (200+ indicators across 8 categories) rather than expert annotation—a deliberate trade-off favoring reproducibility over precision, whose consequences we quantify in the field overview. Hypothesis scoring relies on LLM-extracted assertions operating on abstracts only; claims embedded in method sections, discussion paragraphs, or supplementary materials are not captured, and the fraction of relevant evidence residing in these sections is unknown. The fidelity and limitations of abstract-only extraction are examined in the extraction pipeline section. The hypothesis definitions and domain taxonomy are informed by, but not identical to, the Active Inference Ontology used by Knight et al. (?); future alignment would enable direct comparison with that earlier analysis.

2.8 Principal Contributions

This work makes five contributions:

1. **A multi-source retrieval and deduplication pipeline** for Active Inference literature, using a canonical identifier hierarchy across three academic databases.
2. **A nanopublication-based knowledge graph schema** encoding directed, confidence-scored assertions about eight core hypotheses with full provenance tracking.
3. **A quantitative field overview** characterizing the growth, domain distribution (A/B/C taxonomy), citation topology, and latent topic structure of the Active Inference literature, with specific attention to how recent benchmark results (detailed in the domain analyses) are reshaping the scalability and application landscape.
4. **An LLM-based hypothesis scoring dashboard** that produces differentiated evidence profiles with temporal trend visualization.
5. **A tooling assessment** of the software ecosystem supporting Active Inference research, including the implemented extraction pipeline, existing software (pymdp, SPM, RxInfer.jl), and knowledge graph infrastructure.

The remainder of this paper is organized as follows. The methodology section describes the five-stage pipeline—the central contribution enabling reproducible, automated evidence synthesis—with separate treatments of literature retrieval, LLM-based assertion extraction, bibliometric analysis, the nanopublication-based knowledge graph, and visualization and variable injection. The hypothesis evidence landscape presents quantitative scoring results (RQ3), followed by the field overview with domain-level analysis (RQ1, RQ2), detailed domain analyses, text analytics, and citation network topology. The conclusion addresses limitations and future directions; the discussion provides community recommendations and open questions. Appendix ?? collects mathematical and algorithmic details; Appendix ?? surveys the tooling landscape (RQ4).

3 Methodology: Pipeline Design and Formal Definitions

This section describes the five-stage computational meta-analysis pipeline. Each stage corresponds to a tested, independently executable script that reads upstream outputs and produces structured artifacts. The pipeline extends the systematic literature analysis approach of Knight et al. (?)—which combined manual annotation with ontology-based automated analysis—by substituting manual coding with fully automated, LLM-driven assertion extraction and citation-weighted hypothesis scoring. All code, configuration files, and reproducibility instructions—including a Dockerized execution environment to guarantee dependency isolation—are publicly available in the project repository (https://github.com/ActiveInferenceInstitute/act_inf_metaanalysis).

3.1 Pipeline Overview

The five-stage pipeline is summarized in ??.

Table 1: Five-stage computational meta-analysis pipeline. Each stage corresponds to an independently executable script that reads upstream outputs and produces structured artifacts. Cross-references link to detailed methodology sections.

Stage	Script	Primary Input	Primary Output
1	01_literature_search.py	API queries	corpus.jsonl
2	02_meta_analysis_pipeline.py	corpus.jsonl	Classification, temporal, TF-IDF, NMF, citation network
3	03_build_knowledge_graph.py	corpus.jsonl	nanopublications.jsonl, nanopublications.trig, scor
4	04_generate_figures.py	All Stage 2–3 JSONs	16 publication-ready PNGs
5	05_inject_variables.py	All output JSONs	Rendered manuscript Markdown

Scripts act as thin orchestrators that import methods from tested library modules and handle file I/O. All computation resides in the `src/` packages; no analysis logic is embedded in scripts. End-to-end pipeline execution completes in under one hour on commodity hardware (excluding LLM extraction, which depends on model size and inference backend); all stochastic components use fixed random seeds for deterministic reproduction.

3.2 Reproducible Build Infrastructure

The five-stage analysis pipeline described above is embedded within `template/` (??), an open-source Infrastructure-as-Code system for computational research that turns a full research compendium—code, data, tests, manuscript, and provenance—into a single, version-controlled, deterministically buildable repository with an enforced, test-gated publication pipeline. `template/` applies the principle of Infrastructure as Code to the research lifecycle, making the manuscript, test suite, and provenance chain independently verifiable. The system operationalizes FAIR4RS principles (?) and supply-chain-style provenance for manuscripts, targeting structural causes of the reproducibility crisis: fragmented workflows across LaTeX, notebooks, and ad-hoc scripts, lack of end-to-end testing, and no binding between code, data, figures, and the final PDF.

The system employs a Two-Layer Architecture: a globally shared *infrastructure layer* (12 subpackages, approximately 150 Python modules) provides generic services—logging, rendering, validation, steganographic watermarking, reporting, and LLM integration—while self-contained *project workspaces* (including the present meta-analysis) carry their own `manuscript/`, `scripts/`, `src/`, `tests/`, `data/`, and `output/` directories, discovered purely by filesystem convention. An eight-stage build pipeline enforces an ordered sequence from environment setup through test execution (at least 90% coverage for project code, at least 60% for shared infrastructure), analysis execution, PDF rendering (Pandoc to LaTeX to XeLaTeX with biber), output validation, LLM review, and executive reporting. A Zero-Mock testing policy requires all tests to exercise real filesystem operations, real subprocess calls, and real computation—no `unittest.mock` doubles—making test adequacy a publication gate rather than a best-effort guideline. Cryptographic provenance is embedded in every PDF via SHA-256 hash manifests, PDF metadata injection, and optional QR codes linking back to the repository. A Documentation Duality standard equips every directory with both human-readable `README.md` and machine-readable `AGENTS.md` files, while each infrastructure module carries a `SKILL.md` skill descriptor aligned with the Model Context Protocol, enabling AI agents to locate and invoke module capabilities without hallucinating API signatures. The `template/` framework and this meta-analysis project are available under the Apache 2.0 License at https://github.com/ActiveInferenceInstitute/act_inf_metaanalysis.

3.3 Stage 1: Multi-Source Literature Retrieval and Deduplication

We retrieve papers from three complementary academic databases to maximize coverage and enable cross-source deduplication. The retrieval window begins at 2005, encompassing the period when Energy-Based Model and variational Bayesian research (??) provided mathematical precursors to what Friston formalized as the Free Energy Principle in 2006 (?); this inclusive start captures historical lineage and cross-disciplinary convergence that a later cutoff would exclude.

arXiv. We query the arXiv Atom API using five phrase-matched searches by default: `all:"active inference"`, `all:"free energy principle"`, `all:"predictive coding" AND all:"free energy"`, `all:"expected free energy"`, and `all:"variational free energy" AND all:"inference"`. The `all:` prefix searches titles, abstracts, and full text; phrase matching reduces contamination from unrelated physics papers that mention “free energy” in thermodynamic contexts. Additional Energy-Based Model queries (`all:"energy-based model" AND all:"free energy"`, `all:"Helmholtz machine" AND all:"inference"`, `all:"Boltzmann machine" AND all:"free energy"`, `all:"contrastive divergence" AND all:"generative model"`) are available via the `arxiv_queries` list in `config.yaml` for researchers wishing to capture adjacent EBM literature at the intersection of energy-based generative modeling and variational inference (?).

Semantic Scholar. We query the Semantic Scholar Graph API (?) with the same terms. Semantic Scholar provides citation graphs, abstract embeddings, and links to published versions. Retry logic with exponential backoff handles rate limiting.

OpenAlex. We query OpenAlex (?) to capture journal-published work that may not appear on arXiv, including clinical studies and neuroscience experiments in domain-specific venues. The `referenced_works` field populates citation links for each paper.

3.3.1 Canonical Identifier Deduplication

After retrieval, papers are assigned a canonical identifier using the priority scheme: DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > title hash. When the same paper appears in multiple sources, the record with the highest metadata completeness is retained. For each incoming paper, the two records are compared on metadata completeness—defined as the count of non-empty optional attributes across the full Paper record (abstract, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references, publication date, PDF URL, open-access flag, and author list). The pipeline retains the richer record; in the event of a tie, the incumbent is preserved. This “merge-on-add” strategy aggregates the richest available metadata without requiring an expensive downstream reconciliation pass. Deduplication produces $N = 817$ unique papers spanning 2005–2026.

3.3.2 Relevance Filtering and Curation

After deduplication, a **relevance filter** removes papers whose titles and abstracts lack any core Active Inference terminology (e.g., `active inference`, `'free energy principle'`, `'variational free energy'`), eliminating off-topic results introduced by broad keyword overlap across heterogeneous databases. We acknowledge that this retrieval strategy yields limited bibliographic depth, functioning as a representative snapshot rather than an exhaustive census of the literature.

We emphasize that this process relies on keyword search strategies across divergent APIs. In any complex research field, there is no single optimal word or threshold for definitive inclusion or exclusion. Different information sources and repositories yield differing schemas and representations, introducing both false positives (e.g., machine learning papers that mention “free energy” in a purely thermodynamic context, or bioinformatics tools whose names overlap with AIF terminology) and false negatives (e.g., predictive coding studies that avoid the phrase “free energy principle” entirely, or agent-based modeling papers that implement functionally equivalent algorithms under different labels). The keyword lists in `config.yaml` document all search terms explicitly to enable systematic replication and refinement.

Consequently, this pipeline is not intended to produce a static, “golden” list of canonical papers. Rather, it is designed as an open-source software package that can be modularly updated and versioned. Researchers can configure the pipeline to operate on custom literature bibliographies curated for specific relevance criteria through time, treating the initial query-based retrieval as a programmatic starting point rather than an absolute boundary. For example, adding a ninth domain category (e.g., “D: Education”) requires only adding a keyword list to the `subfield_keyword_s` section of `config.yaml`—no source code modification is needed.

3.4 LLM-Based Assertion Extraction: Prompt Design, Error Taxonomy, and Validation

This supplementary section documents the implementation specifics of the LLM-based assertion extraction pipeline.

3.4.1 Relationship to Prior Approaches

The closest prior effort is the systematic literature analysis of Knight, Cordes, and Friedman (?), which used human annotators to manually code structural, visual, and mathematical features of FEP and Active Inference publications. Their work operated at the scale of hundreds of annotated papers and employed terms from the Active Inference Institute’s Active Inference Ontology for automated text analysis. Our pipeline replaces the manual coding step with LLM-based assertion extraction, enabling scalable processing of the full corpus ($N = 817$ papers) at the cost of exchanging human-verified precision for machine-generated assessments that require post-hoc validation. This trade-off is characteristic of the broader LLM-based scientific extraction landscape: recent benchmarking confirms that even state-of-the-art modular extraction architectures fall short of production-level precision—particularly on tasks requiring exhaustive retrieval and aggregation of multiple values from long documents—validating our design choice to retain human review pathways alongside automated extraction.

Table 2: Comparison of annotation approaches: Knight et al. (2022) manual coding versus this work’s automated LLM-based extraction pipeline. Key trade-offs between human-verified precision and machine-generated scalability are highlighted.

Dimension	Knight et al. (2022)	This work
Scale	Hundreds of papers	817 papers
Annotation	Manual (structural/visual/math features)	Automated (LLM hypothesis assessment)
Ontology	Active Inference Ontology terms	8 standard hypotheses
Output	Annotated features + term frequencies	Nanopublications + knowledge graph
Reproducibility	Annotator-dependent	Deterministic (given model + seed)
Precision	High (human-verified)	Medium (requires validation)

3.4.1.1 Positioning in the LLM-Based Review Landscape Our pipeline operates within a rapidly maturing ecosystem of LLM-powered literature analysis tools. Multi-agent architectures such as LitLLM decompose the review process into specialized sub-agents (planner, identifier, extractor, compiler), while ensemble approaches aggregate outputs from multiple LLMs via weighted voting to improve reliability. Our work differs from these tools in three respects: (1) we target *hypothesis-level evidence scoring* rather than inclusion/exclusion screening; (2) we produce structured nanopublications rather than narrative summaries; and (3) we are only analyzing abstracts for claims. This deliberate trade-off enables corpus-scale processing ($N = 817$) but fundamentally misses fine-grained claims embedded in method sections or discussion paragraphs. Full-text processing could improve extraction recall, particularly for hypotheses with small evidence bases (H6 Clinical Utility, H7 Morphogenesis).

3.4.2 The Eight Tracked Hypotheses

Our analysis tracks the evolving evidence base for eight distinct claims within the Active Inference literature, spanning theoretical universality to applied clinical utility:

1. **H1: FEP Universality (Theoretical).** The Free Energy Principle applies universally to all self-organizing systems.
2. **H2: AIF Optimality (Computational).** Active Inference agents achieve optimal decision-making under uncertainty.
3. **H3: Markov Blanket Realism (Philosophical).** Markov blankets correspond to real physical boundaries.
4. **H4: Predictive Coding (Empirical).** Cortical hierarchies minimize prediction errors via predictive coding.
5. **H5: Scalability (Computational).** Active Inference scales to complex, high-dimensional environments.
6. **H6: Clinical Utility (Applied).** Active Inference provides clinically useful models of psychiatric conditions.
7. **H7: Morphogenesis (Biological).** The FEP explains morphogenetic and developmental processes.
8. **H8: Language AIF (Applied).** Active Inference provides a viable framework for language processing.

3.4.3 Prompt Engineering and Schema Design

The structured prompt is designed to minimize parsing failures and maximize assessment quality:

1. **Explicit JSON schema.** The prompt specifies the exact output schema—field names, allowed direction values, and the numeric confidence range—reducing the LLM’s tendency to generate free-form text or ad hoc structures.
2. **Hypothesis definitions in-context.** All eight definitions are included verbatim, ensuring the LLM assesses relevance from the provided context rather than relying on parametric knowledge that may be stale.
3. **Reasoning field.** Each assessment includes a natural-language reasoning string, providing an audit trail for human reviewers and enabling systematic analysis of error patterns.
4. **Irrelevant filtering.** An explicit “irrelevant” direction allows the LLM to mark hypotheses that a paper does not address, avoiding forced spurious assessments.

3.4.3.1 Prompt Template

The extraction prompt follows a two-part structure (system + user):

SYSTEM: You are a scientific literature analyst specializing in the Free Energy Principle and Active Inference. Assess the relevance of the given paper to each hypothesis. Return a JSON array.

USER:

Paper: {title}

Abstract: {abstract}

Hypotheses:

H1: FEP Universality - {description}

H2: AIF Optimality - {description}

...

H8: Language AIF - {description}

For each hypothesis, return:

```
{
  "hypothesis_id": "H1",
  "direction": "supports|contradicts|neutral|irrelevant",
  "confidence": 0.0-1.0,
  "reasoning": "..."
}
```

The extraction module (`src/knowledge_graph/llm_extraction.py`) includes configurable retry logic with exponential backoff, JSON parsing with handling of markdown code fences and extraneous text, confidence clamping, and validation against the hypothesis ID set. The default model is `gemma3:4b` on a local Ollama instance, configurable via `--llm-model` and `--llm-url` flags.

3.4.4 Failure Modes and Error Recovery

The primary failure modes are documented below.

3.4.4.1 Over-Extraction Bias Preliminary experiments indicated about 15–20% over-extraction. The current $N = 817$ corpus was processed without a validation set; error rates are not quantified for this run. This is the most common error mode and produces false supporting evidence. Over-extraction disproportionately affects broad-scope hypotheses (H1 FEP Universality, H2 AIF Optimality) where most papers in the corpus contain relevant terminology without explicitly endorsing the claim. Narrower hypotheses tied to specific domains (H7 Morphogenesis, H8 Language AIF) show lower over-extraction rates because their vocabulary is more distinctive. This systematic bias inflates support counts for broad hypotheses, and we caution against interpreting absolute scores for H1 and H2 without accounting for this effect.

3.4.4.2 Direction Misclassification The LLM misclassifies a contradicting claim as supporting, or vice versa. Rarer but more consequential, as it directly inverts the evidence signal. Most common for papers that discuss limitations while ultimately endorsing a hypothesis.

3.4.4.3 Confidence Calibration Constraints The model occasionally assigns high confidence to assessments where the underlying evidence is ambiguous. Reliable confidence calibration remains an open problem for zero-shot LLM applications, motivating the multi-tiered validation protocols described below.

3.4.4.4 Progressive JSON Parsing Recovery To mitigate formatting inconsistencies, the module implements a progressive parsing pipeline to recover malformed LLM outputs:

1. **Direct parse:** Attempt `json.loads()` on the raw response.
2. **Strip code fences:** Remove Markdown ````json ... ```` wrappers and retry.
3. **Extract JSON array:** Scan for the first `[...]` substring in the response text.

Papers that fail all parsing stages are logged and skipped; their count is reported at pipeline completion.

3.4.5 Validation Methodology

To calibrate the pipeline in the absence of a human-annotated gold set, we run a **rule-based reference-annotator agreement study** on a stratified sample ($n = 200$ assertions; hypothesis $\times$ triage direction $\times$ year bin). Two *deterministic keyword-and-negation rule protocols*—not human annotators—label each sampled abstract; their mutual agreement (Cohen’s $\kappa = 0.554$) measures how stable the rule reference itself is, and the primary rule protocol serves as an independent, fully reproducible reference against which LLM pipeline triage is compared: precision 0.000, recall 0.000, F1 0.000 (positive class “supports”). These figures are a reproducibility floor, not an accuracy against ground truth: the rule reference and the LLM diverge sharply (direction-agreement $\kappa = -0.055$), which is itself the finding—absolute triage labels are unreliable and only relative rankings should be trusted. Both protocols emit three separable layers: source claim text, evidence supply (status and type), and hypothesis triage direction. A human gold-standard annotation remains future work (see the conclusion’s “Human spot-check coverage”).

Table 3: Pipeline-versus-rule-reference agreement metrics (stratified sample). The reference labels are produced by deterministic keyword rules, not human annotators; values are a reproducibility floor, not accuracy against ground truth.

Metric	Value
Sample size	200
Inter-rule <i>kappa</i> (reference stability)	0.554
Reference–pipeline direction <i>kappa</i>	-0.055
Pipeline precision vs. reference (supports)	0.000
Pipeline recall vs. reference (supports)	0.000
Pipeline F1 vs. reference (supports)	0.000
Over-extraction rate	0.765

Error taxonomy rates (over-extraction, direction inversion, triage mismatch) are reported in `output/reports/validation_metrics.json`; the dominant mode is over-extraction (0.765 of sampled rows), where the LLM assigns a hypothesis label the keyword reference treats as irrelevant. Sensitivity analysis across six citation-weight policies yields rank-stability Spearman $\rho = 0.762$ versus the default log-citation weighting, with 18 hypothesis rank-position changes across all alternative policies.

3.4.6 From Assertions to Nanopublications

Each assertion is wrapped in a **nanopublication** (??) that carries a structured provenance block—paper ID, source passage, model ID (`gemma3:4b`), prompt version (`v2.0.4`), processing date, pipeline version, and run ID. Every assertion in the corpus is produced by the three-layer extractor and carries this block; the aggregate model, prompt-version, and processing-date coverage is reported in `output/reports/extraction_provenance_summary.json`.

Nanopublications are persisted **incrementally** during extraction. Every 50 papers (configurable via `--checkpoint-interval`), the pipeline atomically appends newly extracted nanopublications to `nanopublications.jsonl` using a temporary-file-plus-rename strategy that prevents corruption on interruption. Deduplication operates on the composite key (`paper_id, hypothesis_id`): when a paper is re-processed with an improved model, the newer assertion overwrites the stale entry. This merge-on-add design enables iterative model refinement without costly full-corpus re-extraction.

After extraction completes, the full nanopublication set is additionally serialized to **RDF/TriG** format per the nanopublication standard, producing four named graphs per nanopublication (Head, Assertion, Provenance, Publication Info). The TriG output is suitable for publication to the decentralized nanopublication network and archival on data repositories such as Zenodo. The complete RDF schema is specified in the knowledge graph methodology and Appendix ??.

3.5 Stage 2: Bibliometric Analysis

Stage 2 performs four complementary analyses on the deduplicated corpus. All analyses are deterministic given fixed random seeds and operate on the same `corpus.jsonl` input.

3.5.1 Subfield Classification

Each paper is classified into one of eight categories organized across three domains: **A – Core Theory** (A1: quantitative and formal mathematical theory; A2: qualitative philosophy and general FEP theory), **B – Tools & Translation** (algorithms, scaling, and software development), and **C – Application Domains** (C1: neuroscience, C2: robotics, C3: language processing, C4: computational psychiatry, C5: biology and morphogenesis). Classification uses word-boundary-aware keyword matching against curated lists (74+ mathematical indicators, 25+ philosophy terms, 24+ tools terms, and 14–20 terms per application domain—totaling over 200 keywords across 8 categories, all documented in `config.yaml`) applied to titles and abstracts. A priority system ensures that specific application domains (C1–C5, priority 1) take precedence over tools (B, priority 2), formal theory (A1, priority 3), and the broad qualitative philosophy catch-all (A2, priority 4). Within a priority tier, the domain with the most keyword matches wins. A1’s keyword set includes mathematical indicators such as *theorem*, *proof*, *convergence*, *posterior*, *equation*, and *Fokker–Planck*, ensuring that papers with mathematical content are classified as formal theory rather than defaulting to the philosophy category.

3.5.2 Temporal Metrics and Growth-Rate Estimation

We compute temporal publication metrics including year-by-year counts with gap-filling, cumulative totals, 3-year smoothed moving averages, and peak year identification. Field dynamics are estimated via two complementary metrics. The **mean year-over-year growth rate** $\bar{g}$ is the arithmetic mean of annual growth rates for years with non-zero prior-year publications. The **doubling time** $t_d = \ln 2 / \ln(1 + \bar{g})$. The **compound annual growth rate** (CAGR) captures the annualized rate across the full temporal span. Mathematical details are provided in Appendix ??.

3.5.3 Text Analytics

We construct the TF-IDF matrix using tokenization with stopword removal and L2-normalized smoothed term-frequency inverse-document-frequency weighting (?), with a configurable vocabulary size (default: 500 features). We apply non-negative matrix factorization (NMF) to discover latent topics using multiplicative update rules (?). Topic count $k = 5$ was selected via expert-driven assessment. Mathematical details are provided in Appendix ??.

3.5.4 Citation Network Construction

We construct the intra-corpus citation network as a directed graph where nodes are papers and edges represent citation relationships resolved within the corpus. Because identifier formats vary across databases (arXiv IDs, DOIs, Semantic Scholar IDs), only references whose identifiers match a corpus entry contribute edges; the resulting resolution rate (7.4%) represents a lower bound on the true intra-corpus citation density. Network metrics include PageRank centrality, HITS hub and authority scores (?), degree distributions, network density, connected components, and community structure via greedy modularity maximization (?).

3.6 Stage 3: Nanopublication-Based Knowledge Graph

Stage 3 is the methodological core of this work: it transforms unstructured abstracts into a structured, RDF-compatible knowledge graph of scientific evidence. The stage encompasses four tightly coupled operations: LLM-based assertion extraction, nanopublication packaging, knowledge graph construction, and citation-weighted hypothesis scoring.

3.6.1 LLM-Based Assertion Extraction

We extract assertions by prompting a locally hosted LLM (Ollama (?)) to assess each paper’s abstract against eight standard hypotheses. The model receives a structured prompt containing the paper title, abstract, and hypothesis definitions, and returns a JSON array where each element specifies a hypothesis ID, direction (supports, contradicts, neutral, or irrelevant), a confidence score $c \in [0, 1]$, and a reasoning string. Assertions marked “irrelevant” are discarded; confidence values are clamped to $[0, 1]$; and responses are validated against the known hypothesis ID set. Papers lacking abstracts are skipped. Detailed prompt engineering, error taxonomy, and validation methodology are documented in the extraction pipeline section.

3.6.2 Nanopublication Schema and RDF Structure

Each assertion is encoded as a **nanopublication** (??)—a minimal, self-contained, machine-readable unit of scientific evidence. Formally, each nanopublication is a tuple (p, h, d, c) where p is the paper identifier, h the hypothesis identifier, $d \in \{\text{supports, contradicts, neutral}\}$ the direction, and c the confidence. Provenance metadata records the LLM model, UTC timestamp, and paper identifier.

The pipeline serializes nanopublications in two complementary formats:

1. **JSON Lines** (one JSON object per line) for efficient incremental checkpointing. Assertions are saved at configurable intervals (default: every 50 papers), enabling the pipeline to resume from where it left off after interruption without re-processing already-analyzed papers. Deduplication uses the composite key $(paper_id, hypothesis_id)$; re-runs with improved models overwrite stale results.
2. **RDF/TriG** per the nanopublication standard (nanopub.net), producing four named graphs per nanopublication:

Table 4: RDF/TriG nanopublication structure. Each nanopublication contains four named graphs encoding the assertion, its provenance, and publication metadata per the nanopublication standard (nanopub.net).

Named Graph	Content	Key Predicates
Head	Links the nanopub resource to its three component graphs	<code>np:hasAssertion</code> , <code>np:hasProvenance</code> , <code>np:hasPublicationInfo</code>
Assertion	The core scientific claim	<code>aif:asserts</code> , <code>aif:supports</code> / <code>aif:contradicts</code>
Provenance	How the assertion was generated	<code>prov:wasGeneratedBy</code> , <code>prov:generatedAtTime</code>
Publication Info	Metadata about the nanopublication itself	<code>dc:created</code> , <code>dc:creator</code> , <code>dc:license</code>

The namespace `http://activeinference.institute/ontology/` (prefix `aif:`) defines all domain predicates; the nanopublication schema (`http://www.nanopub.org/nschema#`, prefix `np:`) provides structural predicates; provenance uses PROV-O (`http://www.w3.org/ns/prov#`); and Dublin Core (`http://purl.org/dc/terms/`) provides publication metadata. The TriG output is suitable for publication to the decentralized nanopublication network and aligns with FAIR data principles: **F**indable via URI-based identification, **A**ccessible via standard RDF protocols, **I**nteroperable through W3C-standard serialization, and **R**eusable with explicit provenance and CC0 licensing.

3.6.3 Knowledge Graph Construction

The knowledge graph is an RDF-compatible directed graph with three node types: **paper nodes** (metadata: title, abstract, authors, year, venue, citation count, domain), **assertion nodes** (claim text, direction, hypothesis ID, confidence), and **hypothesis nodes** (the eight standard hypotheses). Edges encode five relations defined in the schema:

- `aif:asserts` — Paper $\rightarrow$ Assertion
- `aif:cites` — Paper $\rightarrow$ Paper
- `aif:belongsTo` — Paper $\rightarrow$ Subfield
- `aif:supports` — Assertion $\rightarrow$ Hypothesis
- `aif:contradicts` — Assertion $\rightarrow$ Hypothesis

The graph is implemented with a dual backend: `rdflib` (?) when available (preferred for semantic web compatibility), with automatic fallback to `networkx.DiGraph` for environments without RDF dependencies. Both backends maintain identical internal indices for efficient paper, assertion, and hypothesis queries.

3.6.4 Citation-Weighted Hypothesis Scoring

For each hypothesis H , we compute a citation-weighted evidence score:

$$\text{score}(H) = \frac{\sum_{a \in S(H)} w(a) - \sum_{a \in C(H)} w(a)}{\sum_{a \in A(H)} w(a)} \quad (1)$$

where $S(H)$, $C(H)$, and $A(H)$ are the sets of supporting, contradicting, and all assertions for H , and the weight function is:

$$w(a) = \log(1 + \text{citations}(a)) \cdot \text{confidence}(a) \quad (2)$$

The logarithmic citation weighting ensures that highly cited papers carry more influence without allowing any single paper to dominate. The score lies in $[-1, 1]$. **Interpretation note:** a score of $+0.7$ indicates that 70% of weighted evidence supports the hypothesis (net of contradictions and normalized by total weighted evidence), *not* that the hypothesis has a 70% probability of being true. Scores are best interpreted as relative rankings across hypotheses and as temporal trajectories within a hypothesis, rather than as absolute probability estimates. Temporal trends are computed by evaluating the cumulative score at each year, using only assertions from papers published up to that year. A full derivation appears in Appendix ??.

3.6.5 Tally-Based Evidence Aggregation

We emphasize that this algorithmic scoring formula constitutes a **tally-based approach** to evidence synthesis: each nanopublication assertion operates as an independent evidential vote, weighted by citation impact and the extraction model’s confidence. The aggregation is linear and additive—supporting and contradicting assertions are summed and differenced without modeling dependencies, correlated evidence, or causal structure among claims. This design choice prioritizes transparency, reproducibility, and computational tractability over statistical sophistication.

The tally-based framing introduces three constraints. First, assertions from methodologically related papers (e.g., iterative publications from a single research group testing the same model) are counted independently, amplifying correlated evidence. To illustrate: if a group publishes three papers (2019, 2021, 2023) reporting successively refined variants of the same predictive coding model, each with high citation counts, the scoring formula counts three independent supporting assertions for H4—even though the underlying empirical evidence is largely overlapping. An evidential diversity index (proposed in the conclusion) would downweight this cluster. Second, the scoring metric treats all assertion sources symmetrically: an assertion from a theoretical review and one from an empirical trial carry equal weight at a given confidence level. Third, temporal scoring tracks *cumulative totals* rather than dynamic probabilistic estimates; the score at year t is the sum of all historical evidence, rather than a decaying posterior that downweights early work.

We embrace these constraints intentionally. The tally-based approach provides a stable, interpretable baseline against which more sophisticated scoring methods can be evaluated. The conclusion describes concrete extensions—including hierarchical Bayesian scoring, causal evidence graphs, and evidential diversity indices that downweight correlated evidence.

3.7 Stages 4–5: Visualization, Variable Injection, and Reproducibility

3.7.1 Stage 4: Visualization

Stage 4 renders 16 publication-ready figures from the analysis outputs of Stages 2 and 3. All figures use the Wong (2011) colorblind-safe palette (?) and enforce a 16-point minimum font size for accessibility compliance. Figures span six categories: field summary and domain distribution (2 figures), growth and temporal dynamics (2 figures), citation network topology (2 figures), hypothesis evidence dashboard and timeline (2 figures), assertion composition (2 figures), and text analytics—word cloud, PCA embeddings, term heatmap, dendrogram, topic-term bars, and co-occurrence matrix (6 figures). The figure generation script reads only JSON outputs and produces only PNG files, enforcing a strict, unidirectional data flow that prevents visualization operations from inadvertently modifying analytical results.

3.7.2 Stage 5: Manuscript Variable Injection

Stage 5 computes dynamic variables from all pipeline outputs and injects them into manuscript Markdown templates via double-brace placeholder substitution of the form `{{<>}}` wrapping a variable name (e.g. the literal token spelled `{{<CORPUS_SIZE>}}` becomes the rendered corpus count). Variables include corpus-level metrics (size, year range, CAGR), per-domain counts and percentages, citation network statistics (nodes, edges, density, components, resolution rate, mean in-degree), hypothesis scores, and figure counts. All formatting (comma thousand separators, escaping) is applied during variable computation, ensuring the manuscript templates remain human-readable while producing publication-ready output. Unrecognized placeholders are preserved with a warning logged, enabling incremental manuscript development ahead of full pipeline execution.

3.7.3 Reproducibility and Test-Driven Validation

The pipeline is deterministic given fixed random seeds and API responses. Test-driven development enforces 90% minimum code coverage on project modules and 60% on shared infrastructure, with real data and computation (no mocking). The test suite validates boundary conditions for hypothesis scoring (all-support $\rightarrow +1$, all-contradict $\rightarrow -1$, balanced $\rightarrow 0$), schema consistency, serialization round-trips, and end-to-end pipeline integrity. Source code, configuration, and outputs are available under CC-BY-4.0.

4 Results

4.1 Hypothesis Evidence Landscape and Temporal Dynamics

The LLM-based extraction pipeline produced a total of 1,671 assertions across the eight tracked hypotheses, drawn from the full corpus of $N = 817$ papers. Before presenting the results, we reiterate the interpretive framework established in the methodology: hypothesis scores are *relative rankings* among hypotheses and *temporal trajectories* within each hypothesis—they are not absolute probability estimates. Publication bias and linguistic asymmetry (§??) inflate all scores toward the positive end, and the tally-based aggregation does not model evidential dependencies. The distribution of assertion types and the resulting citation-weighted scores reveal a differentiated evidence landscape (Figure ??):

Table 5: Citation-weighted hypothesis evidence landscape ($N = 817$ papers, 1,671 total assertions). Scores are computed via (??) and range from -1 (unanimous contradiction) to $+1$ (unanimous support). “Character” summarizes the qualitative evidence profile for each hypothesis.

Hypothesis	Score	Supports	Neutral	Contradicts	Total	Character
H7: Morphogenesis	+0.55	17	2	4	23	Strong consensus
H2: AIF Optimality	+0.68	351	44	32	427	Strong consensus
H4: Predictive Coding	+0.91	132	14	2	148	Strong consensus
H6: Clinical Utility	+0.84	29	7	2	38	Strong consensus
H5: Scalability	+0.73	99	8	2	109	Strong consensus
H8: Language AIF	+0.81	28	2	2	32	Strong support
H3: Markov Blanket Realism	-0.82	5	8	108	121	Moderate, active debate
H1: FEP Universality	+0.13	67	703	3	773	Broad but diffuse

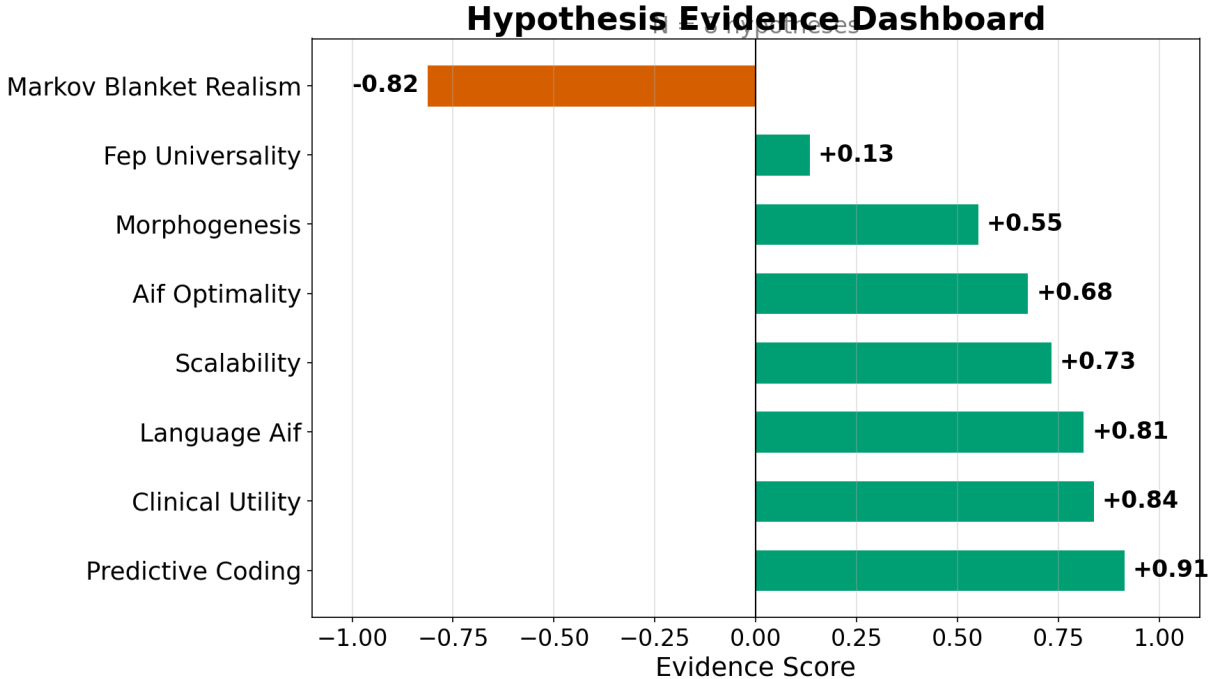


Figure 1: Hypothesis scoring dashboard showing citation-weighted evidence scores ($[-1, +1]$) for the eight tracked hypotheses, sorted descending by consensus strength. Predominantly positive scores reflect both genuine empirical support and systematic positive biases from publication selection and linguistic framing (see §??).

4.1.1 Interpretation of Evidence Profiles

To directly address our core research questions—identifying which claims are robustly supported and which remain contested—we evaluated how the hypothesis-level evidence maps against the critiques introduced in §??. The eight

hypotheses cluster into three tiers, defined by score ranges that emerge from the data rather than being imposed a priori. The **consensus tier** (score > 0.83 ; H7, H2, H4, H6, H5) spans five of the eight hypotheses, revealing a predominantly supportive evidence landscape across domains. H8 (Language AIF) sits at the **boundary of consensus** at score $+0.81$ — above 0.8 but below the more stringent 0.83 line that separates the densely populated upper tier from the rest. Morphogenesis (H7) achieves the maximum score ($+0.55$), though its small evidence base (23 assertions) means unanimity reflects limited assessment scope rather than mature empirical closure. AIF Optimality (H2) holds the second highest score ($+0.68$) despite carrying the largest raw count of contradicting assertions (32): supporting assertions are substantially more highly cited than critical ones, so citation-weighting amplifies the supportive signal—underscoring that citation-weighted scores capture *which* claims the community cites most, not a simple ballot of assertion counts. Predictive coding (H4), the most extensively assessed hypothesis with 148 assertions and a score of $+0.91$, has accumulated overwhelmingly supportive evidence since the 1970s, reflecting the deep empirical grounding of hierarchical prediction error models in neuroscience. This trajectory is consistent with the manual benchmarking results of Knight et al. (?), which similarly identified predictive coding as the most rigorously validated construct in the corpus. Clinical Utility (H6, $+0.84$) and Scalability (H5, $+0.73$) complete the upper consensus tier; H5’s trajectory accelerated sharply after 2017 as deep active inference architectures emerged. The H8 (Language AIF) boundary placement noted above reflects recent breakthroughs coupling active inference to large language models within a still-maturing evidence base.

The **moderate tier** (score 0.5 – 0.8 ; H3) contains a single hypothesis. Markov blanket realism (H3) has the smallest overall evidence base (121 assertions) with a score of -0.82 and 108 contradicting assertions—empirically capturing the ongoing philosophical debate between those who treat Markov blankets as real thermodynamic boundaries (**Friston blankets'**) and those who argue they are purely instrumental statistical tools (Pearl blankets') (?). The moderate score for H3 reflects this active ontological debate: the supporting literature is more highly cited but not by a large margin, and the small total evidence base limits inferential confidence.

The **diffuse tier** (score < 0.5 ; H1) is the most diagnostically informative for understanding the field’s intellectual maturation. FEP universality (H1) generates one of the largest raw evidence bases (773 assertions) yet achieves a score of only $+0.13$ —a striking gap explained by assertion composition: neutral assessments account for 703 of those 773 tallies, while supporting assertions number 67 and contradicting assertions just 3. This neutral plurality—more than either supporting or contradicting tallies—reveals that researchers routinely *invoke* the FEP as conceptual scaffolding without subjecting its universality claim to explicit empirical test. This composition is the quantitative fingerprint of the falsifiability critique leveled by Colombo and Seríes (?): a principle elastic enough to accommodate any self-organizing system without generating predictions that distinguish it from alternatives will naturally accumulate invocations rather than tests, and invocations register as neutral in the extraction pipeline.

4.1.2 Temporal Dynamics of Evidence Accumulation

The cumulative evidence timeline (Figure ??) reveals three temporal patterns. First, **early convergence**: H4 (predictive coding) reached positive territory in the late 1990s following the publication of Rao and Ballard’s foundational predictive coding model (?) and has maintained a high score since, reflecting the mature empirical base in cognitive neuroscience. Second, **recent acceleration**: H5 (scalability) and H6 (clinical utility) show steep upward trends after 2017, tracking the emergence of deep active inference tools and computational psychiatry applications. The H5 trajectory reflects a cumulative body of work culminating in benchmark demonstrations such as AXIOM (?), which showed that object-centric world models under AIF can match state-of-the-art deep RL performance—but the temporal trend was already positive before any single result, and the score captures the aggregate rather than any individual paper. Third, **moderate and stable**: H3 (Markov blanket realism) has maintained a score in the moderate range since 2018, with supporting papers partially offset by targeted philosophical critiques—a pattern consistent with ongoing debate rather than either clear consensus or rejection.

4.1.3 Assertion Composition and Distribution

The per-hypothesis composition of assertions (Figure ??) and the multi-panel summary (Figure ??) provide complementary views of the extraction results.

4.1.4 Limitations of the Current Scoring Approach

4.1.4.1 Publication Bias and Linguistic Asymmetry The predominantly positive scores observed across all eight hypotheses should be interpreted with two systematic caveats.

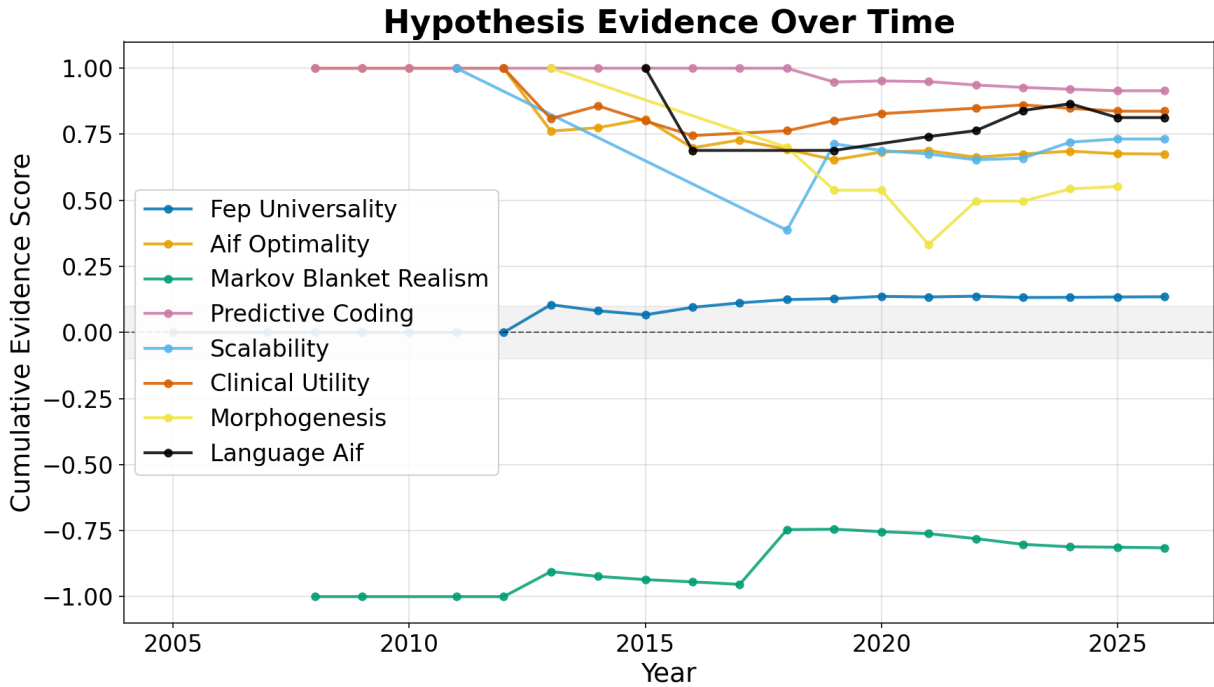


Figure 2: Temporal evolution of cumulative citation-weighted evidence scores by hypothesis (2005–2026). Divergent trajectories around the shaded neutral boundary (± 0.1) reveal which hypotheses are gaining or losing support over time. H4 (predictive coding) stabilized early; H5 (scalability) accelerated post-2017.

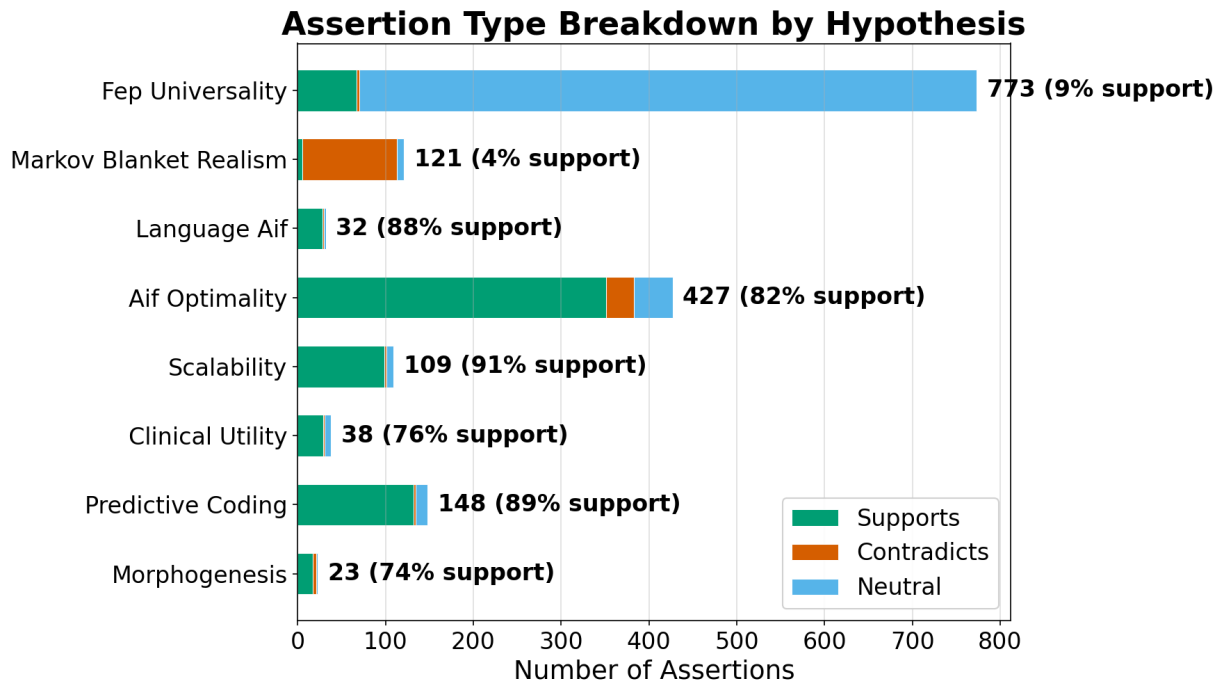


Figure 3: Stacked horizontal bars decomposing per-hypothesis assertions into supports (green), contradicts (red-orange), and neutral (blue) categories ($N = 1,671$ total assertions). Labels show total count and support percentage. The high support fractions are partially attributable to publication bias and affirmative linguistic framing.

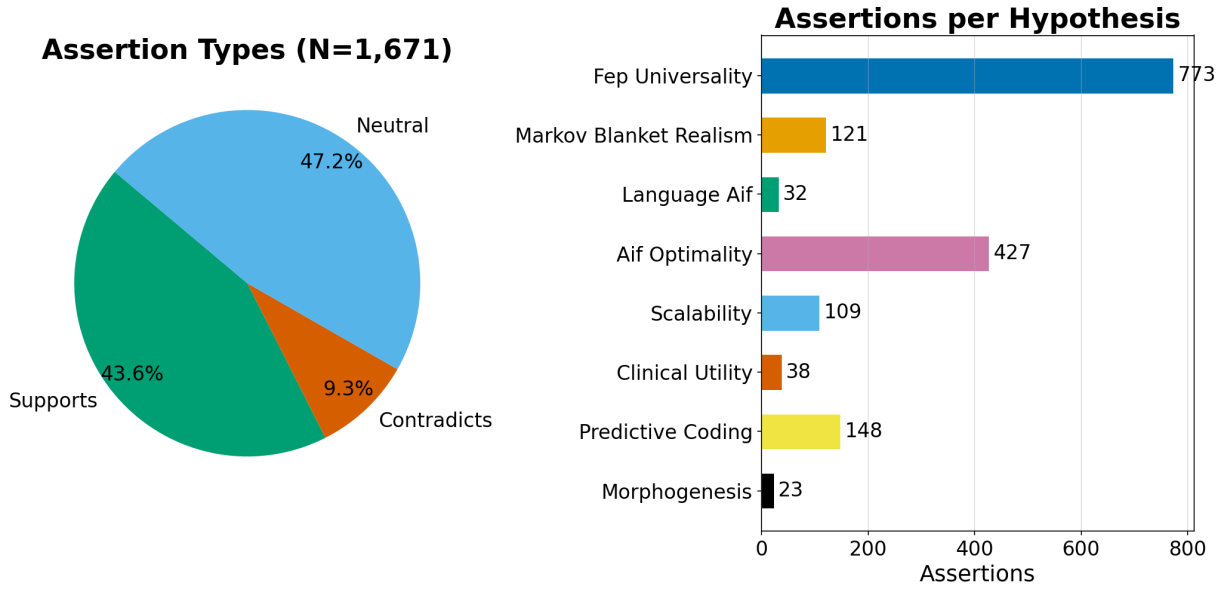


Figure 4: Multi-panel assertion summary: (left) pie chart of overall assertion type distribution showing supports/contradicts/neutral proportions, (right) per-hypothesis assertion counts with palette-coded bars. $N = 1,671$ assertions extracted from 817 papers.

First, **publication bias** systematically inflates supporting evidence. Academic journals preferentially publish positive and confirmatory results (?), meaning that studies finding null or contradictory outcomes for any hypothesis are less likely to appear in the retrievable literature. This *file-drawer effect* is well-documented across scientific disciplines and is expected to disproportionately suppress contradicting assertions in our extraction pipeline. The Active Inference literature is particularly susceptible: as a theoretical framework with strong foundational proponents, papers are more likely to frame results as consistent with the FEP than as challenges to it.

Second, **linguistic asymmetry** in academic writing further skews extraction toward positive classifications. Declarative scholarly claims are inherently phrased affirmatively—authors write “our results support,” “consistent with,” or “extends the prediction of” far more frequently than “our results refute” or “contradicts the claim that.” Because the LLM extraction pipeline operates on abstract text, this linguistic imbalance propagates directly into the assertion distribution. Even papers presenting genuinely mixed evidence tend to frame their abstracts in terms of what *was* found rather than what was not, biasing the extracted direction toward “supports.”

These two effects act in concert: publication bias reduces the number of contradicting papers in the corpus, and linguistic framing reduces the number of contradicting assertions extracted from the papers that do appear. Consequently, the absolute values of hypothesis scores should not be taken as unbiased measures of scientific consensus. The *relative* ordering and temporal *trajectories* of hypothesis scores are more robust indicators, as these biases affect all hypotheses approximately equally.

4.1.5 Methodological Validation and LLM Calibration

The evidence derives from automated LLM-based assertion extraction operating on abstracts only. A stratified rule-based reference-annotator agreement study ($n = 200$) provides a first quantitative, fully reproducible calibration: two deterministic keyword-rule protocols agree at $\kappa = 0.554$ (reference stability), while pipeline triage against the primary rule reference yields precision 0.000, recall 0.000, and F1 0.000, with reference–pipeline direction agreement of only $\kappa = -0.055$. That the LLM and an independent keyword reference diverge this sharply is itself the calibration result: over-extraction (pipeline labels relevant where the keyword reference labels irrelevant) accounts for 0.765 of sampled rows, directly motivating the tempered interpretation of absolute hypothesis scores. Relative rankings and temporal trajectories remain more robust than point estimates. These figures are a reproducibility floor, not accuracy against human ground truth, which remains future work. Full metrics and error taxonomy appear in Tableabout ?? (§??).

4.1.6 Citation-Weighting Sensitivity

Hypothesis ranks under six alternative weight policies remain stable: minimum rank-stability Spearman $\rho = 0.762$ versus the default log-citation policy, with 18 rank inversions across all policies. No hypothesis changes sign under any tested policy; the largest score shifts occur under raw-citation weighting (popularity stress test) for H5 (Scalability) and H8 (Language AIF), reflecting Matthew-effect sensitivity without altering qualitative tier ordering.

4.2 Field Overview: Disciplinary Structure and Growth Dynamics

Annual output in the Active Inference literature rose from 1 papers in 2005, reaching a peak of 123 papers in 2025—a transition from a niche within theoretical neuroscience to a multi-disciplinary research program spanning three primary domains and eight tracked categories. The corpus start of 2005 was chosen to capture Energy-Based Model and variational Bayesian antecedents (??) that preceded the formal introduction of the Free Energy Principle in 2006 (?) and its subsequent full elaboration (?). Our corpus, extracted from arXiv, Semantic Scholar, and OpenAlex and deduplicated to $N = 817$ papers (2005–2026), captures the breadth, tempo, and internal architecture of this expansion (Figure ??).

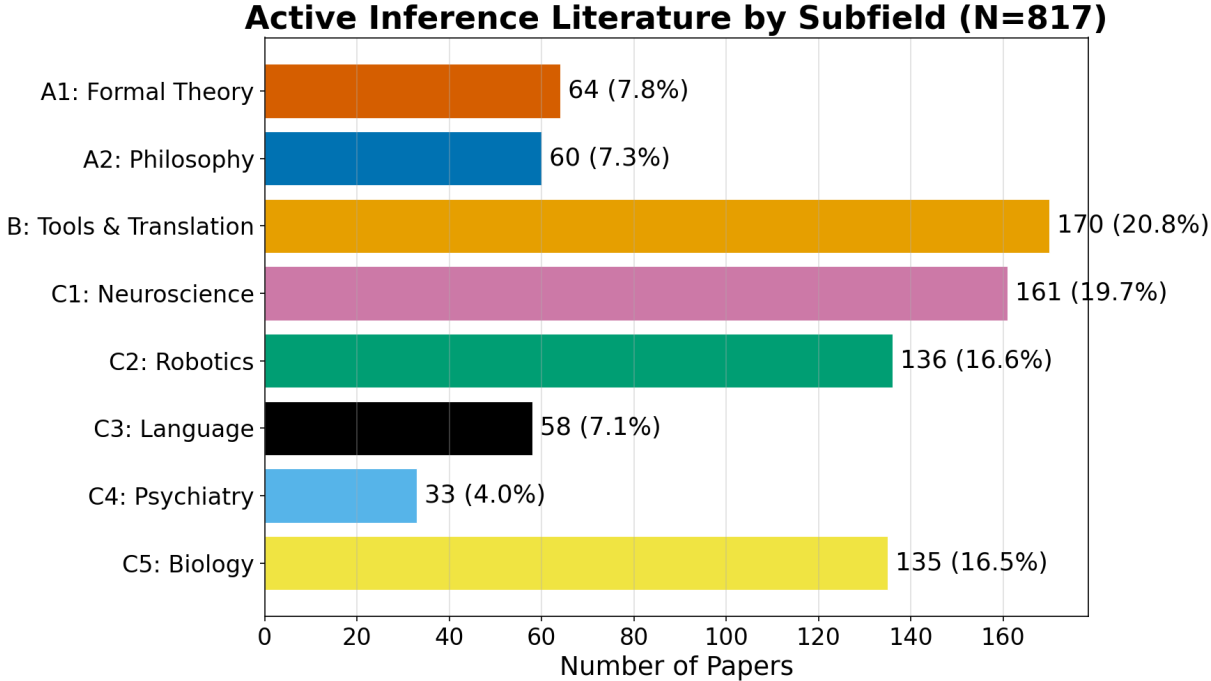


Figure 5: Publication counts by domain ($N = 817$). Application domains (C1–C5) collectively account for the largest share of the corpus; Domain A2 (qualitative philosophy) is the largest single category, reflecting the FEP’s broad theoretical reach.

4.2.1 Corpus-Level Summary

Table 6: Corpus-level summary statistics for the Active Inference literature corpus ($N = 817$), spanning 2005–2026.

Metric	Value
Total papers	817
Year range	2005–2026
Peak year	2025
CAGR	20.36%
Active domains	8 of 8 tracked (A1–A2, B, C1–C5)

The CAGR of 20.36% (measured as the annualised growth rate of yearly publication volume between endpoint years 2005 and 2026) reflects sustained field expansion; annual output accelerated markedly in the late 2010s—from roughly 2018 onward (Figure ??). Sustained high output persisting into subsequent years suggests the field has reached a mature production phase rather than experiencing a transient spike. Citation network metrics are detailed in the dedicated citation network analysis (see the citation network analysis).

4.2.2 Domain Distribution

Keyword-based classification assigns each paper to one of eight categories across three domains:

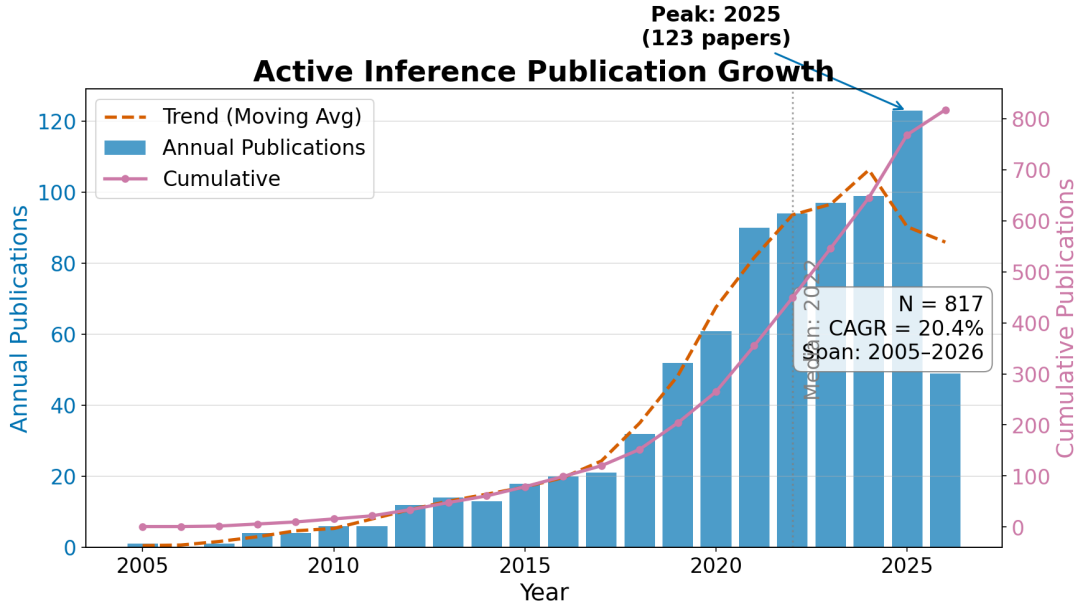


Figure 6: Annual (bars) and cumulative (line) publication counts, 2005–2026 ($N = 817$, $\text{CAGR} = 20.36\%$). Annual output accelerates from the late 2010s (around 2018) to the 2025 peak. Moving average trendline (dashed), peak year, and median year annotated.

Table 7: Domain distribution across three tiers and eight categories ($N = 817$ papers). Classification uses hierarchical keyword matching with priority-based routing to minimize over-assignment to catch-all categories.

Domain	Category	Papers	Percentage
A – Core Theory	A1: Formal Theory	64	7.8%
	A2: Qualitative Philosophy	60	7.3%
B – Tools	B: Tools & Translation	170	20.8%
C – Applications	C1: Neuroscience	161	19.7%
	C2: Robotics	136	16.6%
	C3: Language	58	7.1%
	C4: Psychiatry	33	4.0%
	C5: Biology	135	16.5%

The concentration of papers in A2 (qualitative philosophy and general theory) reflects the broad scope of foundational FEP work (Figure ??). The priority-based classifier mitigates over-assignment by routing papers with mathematical indicators (theorems, proofs, equations, statistical formalism) to A1 before falling back to A2, and by preferring specific application domains (C1–C5) and tools (B) over both core-theory categories. Papers that discuss FEP/AIF conceptually without mathematical formalism or domain-specific vocabulary are correctly assigned to A2. This figure should be read as a *ceiling* on theoretical generality rather than a literal measure of research focus—embedding-based classification would likely redistribute some fraction into more specific categories. That all eight categories are populated, including computational psychiatry (C4) and formal theory (A1), indicates diversification beyond the field’s neuroscience origins.

Detailed characterizations of each domain—including historical context, growth trends, and open problems—are provided in the supplementary domain analyses (see the domain analyses). Latent topic structure, vocabulary analysis, and document embeddings are presented in the text analytics section (see the text analytics section).

4.2.3 Cross-Domain Comparison

Three structural features emerge from the cross-domain comparison (Figure ??). First, no single legacy domain dominates: Domain B (Tools & Translation) accounts for 20.8% of the corpus, followed by C1 (Neuroscience) at 19.7% and C2 (Robotics) at 16.6%. Second, Domain A (Core Theory) aggregates 15.2% collectively (A1 + A2), while the

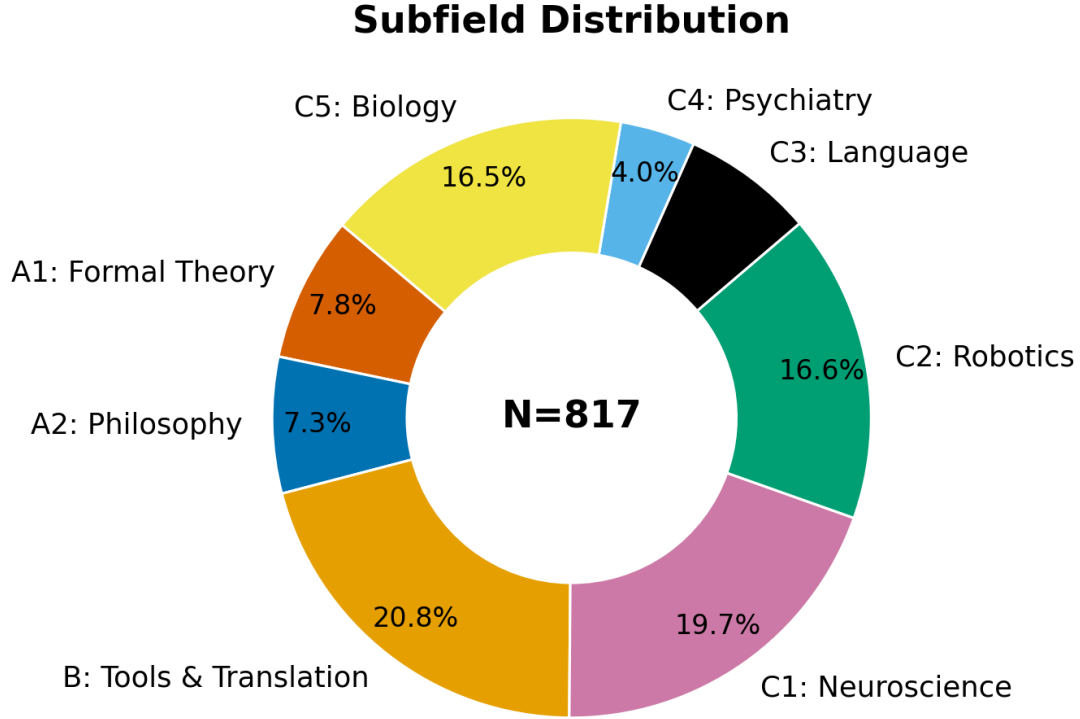


Figure 7: Domain distribution ($N = 817$). Classification uses hierarchical keyword matching against curated lists applied to titles and abstracts, capturing distinct methodological and domain-specific groupings.

Table 8: Cross-domain comparison showing growth trajectories, maturity levels, key challenges, and representative publications for each of the eight tracked categories. Growth trends and maturity assessments are based on temporal publication patterns and evidence base depth.

Domain	Category	Papers	Growth	Maturity	Key Challenge	Rep. Work
A	A1: Formal	64 (7.8%)	Growing	Mature	Math accessibility	(?)
A	A2: Philosophy	60 (7.3%)	Stable	Mature	Catch-all absorption	(?)
B	B: Tools	170 (20.8%)	Rapid	Growing	Deep RL benchmarks	(?)
C	C1: Neuroscience	161 (19.7%)	Stable	Mature	Theory–neuroimaging gap	(?)
C	C2: Robotics	136 (16.6%)	Growing	Growing	Embedded real-time	(?)
C	C3: Language	58 (7.1%)	Emerging	Nascent	NLP model comparison	(?)
C	C4: Psychiatry	33 (4.0%)	Emerging	Nascent	Clinical translation	(?)
C	C5: Biology	135 (16.5%)	Rapid	Nascent	Empirical validation	(?)

emergent application frontiers (C3–C5) exhibit accelerating growth. Third, A1’s 64 papers understate its intellectual influence—the mathematical formalisms developed in A1 shape implementations across all domains.

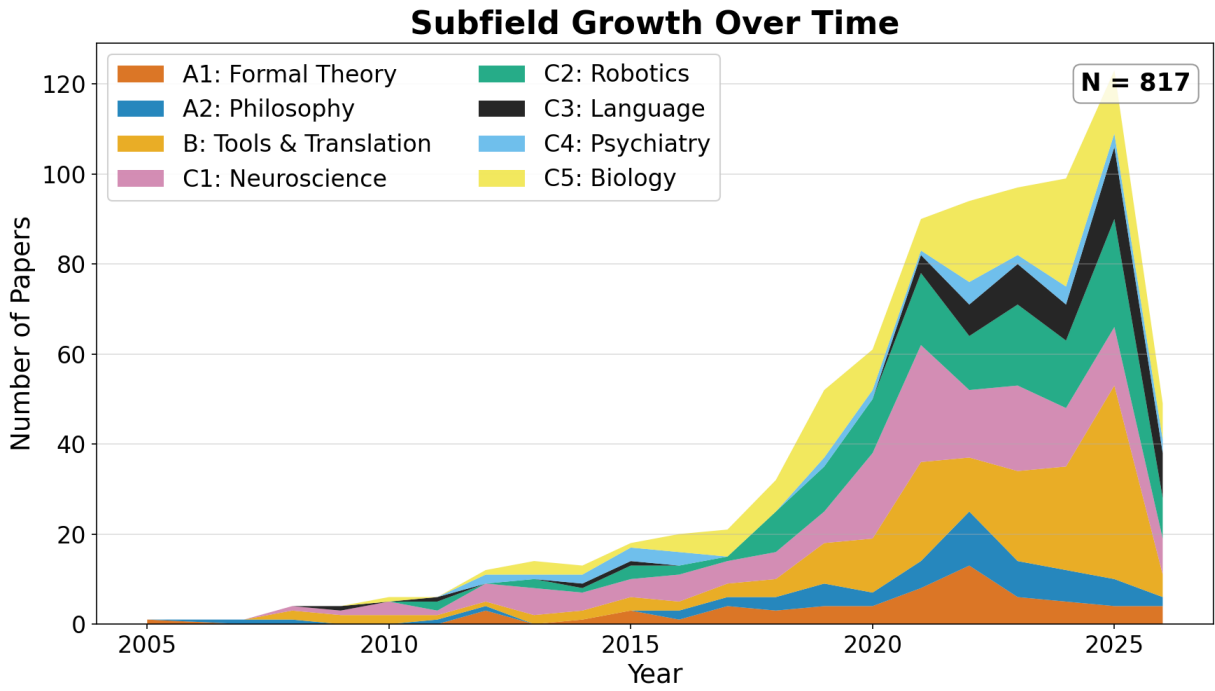


Figure 8: Stacked area chart of publications by domain, 2005–2026 ($N = 817$). A2 (qualitative philosophy) provides a large baseline; application domains C1–C5 show accelerating diversification from 2015 onward.